End-Term Examination

(B.Tech CSEAI/ECEAI CBCS)(SUBJECTIVE TYPE)

(April, 2022)

Subject Code: BAI-101	**Subject: Intelligent Systems**
Time : 3 Hours	**Maximum Marks : 60**

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.		**(4*5=20)**
	(a) Differentiate Data, Information, Knowledge and Intelligence.	
	(b) How data is crucial in machine learning? Why we require pre-processing?	
	(c) When user send a request to open any application using web browser then, what are various steps taken by web application to show result to user.	
	(d) How do you differentiate between automation and intelligent automation? Take one real life example to explain.	
	(e) What is the difference between regression and classification? Use appropriate example.	
	UNIT-1	
Q2.	(i) Formalize the following sentences: 1. "Oshin comes to the party only if Anand and Preet do not come". 2. "Dinesh comes to the party if and only if Oshin comes and Anand doesn't come". 3. "Oshin comes to the party provided that Dinesh doesn't come, but, if Dinesh comes, then Preet doesn't come". 4. "A necessary condition for Anand coming to the party, is that, if Preet and Oshin aren't coming, Dinesh comes". 5. "Anand, Preet and Oshin come to the party if and only if Dinesh doesn't come, but, if neither Anand nor Preet come, then Dinesh comes only if Oshin comes". (ii)Prepare the Truth Table for $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$.	**(6,4)**
Q3.	What is reasoning in AI? Reasoning is classified in AI?	**(10)**
	UNIT-2	
Q4.	A data set is being given to you, your task is to train and test that dataset in python. Write commands to do so. In test and training process what % of dataset is used for testing the data and training the data.	**(10)**
Q5.	What is the difference between classification and clustering? Suppose that you have been given a dataset where you need to apply an appropriate Classification algorithm. Explain various stages of experimentation with the help of a diagram.	**(10)**
	UNIT-3	
Q6.	"For intelligent mobile robots, autonomous navigation is a must-have feature. A robot must do Simultaneous Planning, Localization, and Mapping to travel autonomously in an environment (SPLAM). To successfully handle the dynamic entities in the operating environment, a robot must execute localization, mapping, and planning simultaneously. A robot will not be able to move autonomously in real-world deployment circumstances if none of these tasks are performed" Explain the interaction of Localization, Mapping and Path Planning Activity for Autonomous Navigation.	**(10)**
Q7.	Speech recognition is a field where AI is being used heavily? Discuss the working of few very well-known applications of Speech recognition. Which machine learning model is widely used in that?	**(10)**

UNIT-4

Q8.	"The government's decision to priorities AI for healthcare has prompted more collaboration between the government, technology businesses, and traditional healthcare providers. Several healthcare and tech Companies in collaboration with Government agencies and other private industries and start-ups are working on developing intelligent solutions for early identification of critical disease." So, according to you how AI will help in improving the medical facilities in India, what are various benefits and challenges, stakeholders will face while implementing it.	(10)
Q9.	" In recent years Tesla, Apple, etc are working on self-driving cars so that AI can be implement in car to make our life easier. " Now if a task is being given to you to implement this autonomous vehicle in India, then is it feasible in India to implement it or not, justify your answer with suitable examples.	(10)